

Po-Ju (Austin) Chao

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EDUCATION

Carnegie Mellon University

Master of Science in Mechanical Engineering

National Central University

Bachelor of Science in Mechanical Engineering

Pittsburgh, PA

Expected Completion: Jun 2028

Taoyuan, Taiwan

GPA: 3.75

WORK EXPERIENCE

Industrial Technology Research Institute - Applied Tech Research Organization

Robotics Software Engineer Intern

Hsinchu, Taiwan

Jan 2025 – Jun 2026

Quadruped Locomotion Policy Development

- Developed a PPO-based locomotion policy for a custom quadruped by reward engineering and domain randomization, increasing mean episode reward by 35% and reducing velocity-tracking error by 64%.
- Bridged the sim-to-real gap by learning an actuator model from hardware data and incorporating it into the training simulator, enabling 50 Hz real-time policy control and reducing sim-to-real joint-trajectory error by 45%.

Panda Robot System Development

- Developed a unified motor-control interface for a custom 27-DoF panda robot, abstracting vendor-specific actuators over CAN bus and RS-485 while sustaining 100 Hz state and command updates across all joints.
- Designed a multi-stage, collision-aware eating planner with MoveIt Task Constructor and implemented a 100 Hz cubic-Hermite trajectory executor, reducing the full behavior duration by 78% from 63 s to 14 s.

Visual Perception System Development

- Integrated optimized YOLO models on Jetson Orin Nano using ONNX Runtime GPU, resolving CUDA-Ultralytics compatibility issues and reducing model-inference latency from over 500 ms to 30 ms.

RESEARCH PROJECT

Autonomous Exploration and SLAM Mobile Robot - National Central University

- Developed autonomous navigation stack for differential-drive robot integrating ROS2, 2D LiDAR, IMU, and wheel encoders across SLAM, localization, and path planning, sustaining 90%+ autonomous runs.
- Implemented LiDAR odometry with scan gradient alignment, reducing localization drift by 45% and effectively compensating for IMU and encoder bias during long-term operation.
- Engineered a frontier-based autonomous exploration algorithm in ROS2/Nav2 with custom frontier clustering and filtering, reducing total exploration distance by 25% versus naive nearest-frontier selection.

COMPETITION

Tokyo Electron Limited Robotics Contest - Team Captain & Lead Engineer

Mar 2024 – Dec 2024

- Led design and build of autonomous frisbee-shooting robot integrating feeding, aiming, and launching subsystems; ranked Top 7 nationally, earning Finalist and Capstone Outstanding Project Awards
- Built dual-Arduino Mega architecture splitting chassis and launcher control, with Ziegler-Nichols-tuned PID loops and IMU attitude correction, achieving <200ms motor response
- Engineered stereo-vision targeting backed by regulated buck-converter power and emergency-shutdown safety circuitry, raising shooting accuracy from 60% to 85% while maintaining $\pm 3^\circ$ angular error
- Improved communication reliability by integrating a high-interference-resistant RC controller with signal filtering, extending stable control range beyond 16m across high-EMI, multi-robot competition environments

SKILLS

Programming: C/C++, Python, MATLAB/Simulink

Robot Learning: PyTorch, reinforcement learning, VLM, PPO, NVIDIA Isaac Lab, Isaac Sim, MuJoCo, sim-to-real,

Robotics & Perception: ROS 2, MoveIt 2, Nav2, ORB-SLAM3, OpenCV, YOLO, ICP, sensor fusion

Systems & Deployment: Linux, Git, Docker, CUDA, TensorRT, ONNX Runtime, Jetson Orin, STM32, CAN bus

Controls: PID, LQR, MPC, SMC, EKF